Assignment 2: Convolution

Comparing Training Sample Size and Pretrained Convolutional Neural Networks

Course: BA-64061-002 — Advanced Machine Learning
Semester: Fall 2026

1. Introduction

Convolutional neural networks (CNNs) are commonly used for image classification because they can recognize spatial patterns such as edges, textures, shapes, and more complex visual features. Unlike traditional densely connected neural networks, convolutional networks use filters to identify patterns in local regions of images. As information moves through the network, successive layers learn increasingly complex features.

One challenge when developing CNNs is determining how much training data is necessary to achieve reliable predictions. Smaller datasets may limit a network's ability to generalize to new images, while larger datasets require additional computational resources. Techniques such as data augmentation and dropout can reduce overfitting, but another approach is to use a pretrained convolutional network that has already learned useful image features.

The purpose of this assignment was to investigate the relationship between training sample size and model performance by comparing CNNs trained from scratch with pretrained CNNs. Using images of cats and dogs, six primary experiments were conducted with training samples of approximately 1,000, 2,000, and 10,000 images.

The primary objective was to determine whether increasing sample size improves classification accuracy and whether a pretrained network can achieve comparable or better results using fewer training images.

2. Data and Methodology

2.1 Dataset Preparation

The experiments used the Cats vs. Dogs image classification dataset discussed in Chapter 8 of *Deep Learning with Python* by François Chollet.

The initial dataset was provided through the course materials and contained 2,000 training images, 1,000 validation images, and 1,000 test images. The original images were organized into separate folders for cats and dogs.

For the first experiment, 500 cat images and 500 dog images were randomly selected for training. The validation and test sets were each reduced to 500 images, with 250 cats and 250 dogs.

The second experiment increased the training sample to 2,000 images, maintaining the same 500-image validation and test sets.

For the third experiment, a larger Cats and Dogs dataset was downloaded from Microsoft's publicly available dataset archive. Images matching the existing evaluation images by exact file hash were excluded. Four images incompatible with TensorFlow decoding were subsequently removed, resulting in a training sample of 9,996 images.

All six primary experiments used the same previously selected validation and test datasets to maintain consistency when comparing model performance.

Images were resized to 180 × 180 pixels and processed in batches of 32 using TensorFlow and Keras. The experiments were conducted in Google Colab.

2.2 Convolutional Network Trained From Scratch

The first three primary experiments used a CNN trained from randomly initialized weights. The network consisted of five convolutional layers with increasing filter counts of 32, 64, 128, 256, and 256. Max-pooling layers reduced the spatial dimensions as features progressed through the network.

A flattening layer converted the final feature maps into a representation suitable for binary classification. The output layer contained one neuron with a sigmoid activation function to distinguish between cats and dogs.

Data augmentation was incorporated into the network to reduce overfitting. The transformations included random horizontal flipping, rotation, and zoom. A dropout rate of 0.5 was also used to reduce reliance on individual neurons during training.

The models used binary cross-entropy loss and the RMSprop optimizer with a learning rate of 0.0001. Each model was allowed to train for a maximum of 30 epochs. Early stopping monitored validation loss and restored the best-performing weights when training stopped.

The principal experiments maintained the same model architecture and training settings while varying the number of training images.

Additional exploratory experiments were conducted using 1,500 training images. One incorporated batch normalization and global average pooling, while another returned to the original architecture. These experiments provided supplementary information but were not included in the six primary comparisons.

2.3 Pretrained VGG16 Network

The next three experiments used VGG16, a convolutional neural network previously trained on the ImageNet dataset.

Rather than training every convolutional layer from scratch, the pretrained VGG16 convolutional base was used for feature extraction. Its weights were frozen, allowing the network to retain previously learned visual representations.

A new classification head was added to perform binary classification. This consisted of a flattening layer, a dense layer with 256 neurons and ReLU activation, dropout of 0.5, and a sigmoid output layer.

The pretrained experiments also included random horizontal flipping, rotation, and zoom. VGG16-specific preprocessing was applied to the image inputs.

The pretrained models used the RMSprop optimizer with a learning rate of 0.0001, binary cross-entropy loss, model checkpointing, and early stopping.

The three pretrained experiments used training samples of 1,000, 2,000, and 9,996 images.

Although Chapter 8 also discusses fine-tuning pretrained convolutional layers, these experiments focused on feature extraction with the pretrained convolutional base frozen.

3. Experimental Results

The primary performance measurements were validation accuracy, validation loss, test accuracy, and test loss. The best epoch for each model was identified using the lowest validation loss.

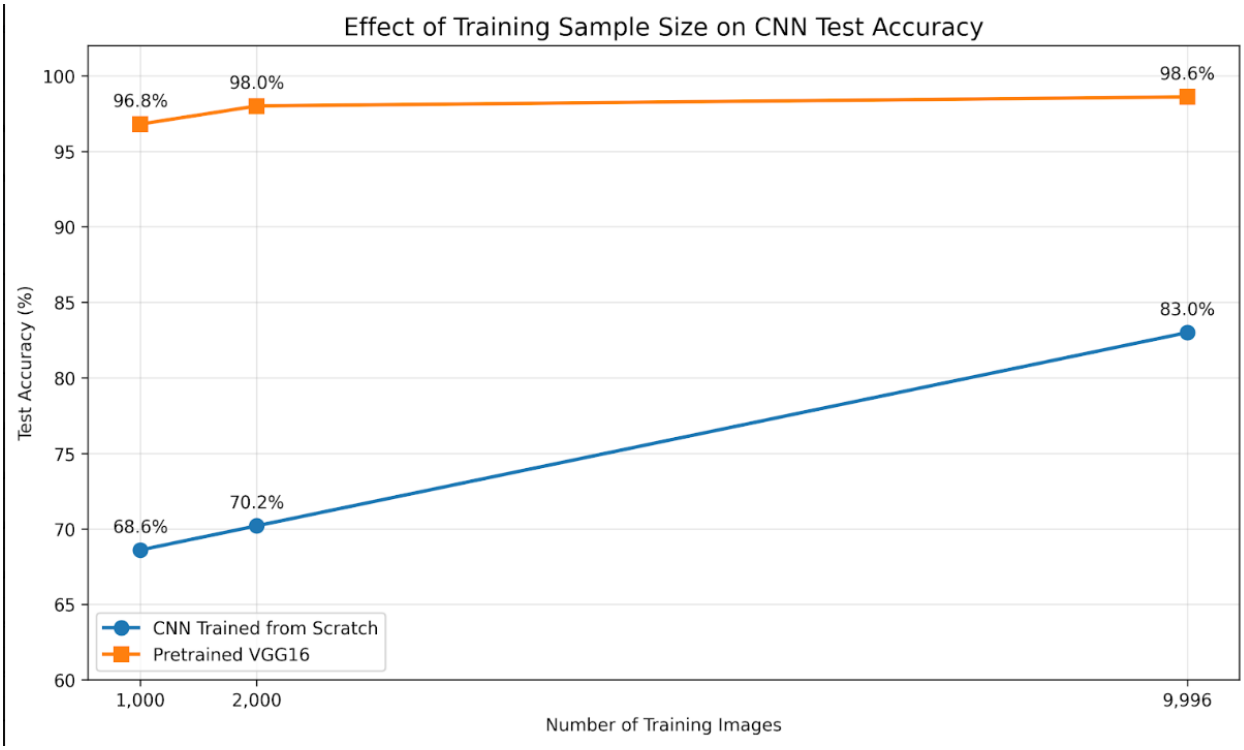
Table 1. Comparison of CNN Performance by Model Type and Training Sample Size

Model	Training Images	Best Epoch	Validation Accuracy	Validation Loss	Test Accuracy	Test Loss
CNN from scratch	1,000	28	64.60%	0.6236	68.60%	0.6196
CNN from scratch	2,000	30	69.80%	0.5509	70.20%	0.5467
CNN from scratch	9,996	30	84.20%	0.3373	83.00%	0.3714

Pretrained VGG16	1,000	10	96.80%	0.2395	96.80%	0.2339
Pretrained VGG16	2,000	7	98.00%	0.1522	98.00%	0.2343
Pretrained VGG16	9,996	12	98.60%	0.0953	98.60%	0.1072

Note. Validation accuracy is reported at the epoch with the lowest validation loss. All experiments used 500 validation images and 500 test images.

Figure 1. Effect of Training Sample Size on CNN Test Accuracy



3.1 Performance of CNNs Trained From Scratch

The results demonstrated that increasing the training sample size improved the performance of the CNN trained from scratch.

With 1,000 training images, the model achieved 68.60% test accuracy. Increasing the training sample to 2,000 images improved test accuracy to 70.20%, representing an increase of 1.6 percentage points.

The most substantial improvement occurred when the training sample was increased to 9,996 images. Test accuracy increased to 83.00%, an improvement of 12.8 percentage points over the 2,000-image experiment and 14.4 percentage points over the initial 1,000-image experiment.

Validation loss also decreased from 0.6236 with 1,000 training images to 0.3373 with 9,996 training images.

These results suggest that increasing training sample size improved the network's ability to recognize cats and dogs in images not used for training.

The largest scratch model achieved its lowest validation loss at epoch 30, the maximum number of epochs permitted in the experiment. This suggests that additional training could be explored, although the present results do not establish whether more epochs would improve performance.

3.2 Performance of Pretrained VGG16 Networks

The pretrained VGG16 models achieved substantially higher accuracy across all three training sample sizes.

With only 1,000 training images, VGG16 achieved 96.80% test accuracy. This was 28.2 percentage points higher than the CNN trained from scratch with the same sample size.

Increasing the sample to 2,000 images improved VGG16's test accuracy to 98.00%. The final experiment, using 9,996 training images, achieved 98.60% test accuracy.

Although the pretrained network continued to improve as training sample size increased, the improvements were considerably smaller than those observed when training the CNN from scratch.

The pretrained models also reached their lowest validation losses in fewer epochs than the corresponding scratch models. The three VGG16 models achieved their best results at epochs 10, 7, and 12, compared with epochs 28, 30, and 30 for the scratch CNNs.

These findings illustrate the advantage of using previously learned features rather than attempting to learn all relevant image patterns from a relatively small training dataset.

3.3 Comparison of Sample Size and Network Choice

The most significant finding was that a pretrained network using only 1,000 training images outperformed a CNN trained from scratch using nearly 10,000 images.

The pretrained VGG16 model achieved 96.80% test accuracy with 1,000 images, compared with 83.00% for the largest CNN trained from scratch.

This suggests that the choice of network architecture and use of pretrained weights can be more important than simply increasing the number of training examples.

When training from scratch, increasing sample size produced substantial improvements because the network needed to learn useful image representations directly from the available training data.

In contrast, VGG16 already contained visual features learned from ImageNet. The new classifier could use these representations to distinguish between cats and dogs without needing to relearn basic visual patterns.

The results also suggest diminishing improvements in classification accuracy as pretrained model performance approaches 100%. VGG16 improved from 96.80% to 98.60% as training samples increased from 1,000 to 9,996 images, compared with the much larger improvement observed for the scratch CNN.

4. Discussion

The experimental findings were consistent with the concepts presented in Chapter 8 of *Deep Learning with Python*.

Convolutional networks are designed to recognize spatial features within images. However, a network trained from scratch must learn these representations using the training examples available for the specific task.

When the training sample is relatively small, the model has fewer opportunities to observe differences in image composition, lighting, orientation, background, and other characteristics. Data augmentation partially addresses this limitation by exposing the model to modified versions of existing training images.

Dropout also helps discourage the model from depending too heavily on particular learned representations.

Despite these measures, the initial scratch CNN achieved relatively modest prediction accuracy. Increasing the amount of original training data provided a more substantial improvement.

The pretrained VGG16 model demonstrated a different approach. By transferring visual representations learned from a much larger dataset, it achieved strong classification performance with considerably fewer task-specific images.

This is particularly relevant in situations where collecting or labeling large amounts of data is expensive or impractical.

The experiments also demonstrate that model selection should consider more than final accuracy. Training requirements, available data, computational resources, and the extent to which pretrained features are relevant to the problem can all influence the most appropriate modeling approach.

5. Limitations

Several limitations should be considered when interpreting the results.

Dataset consistency. The initial experiments used the smaller course dataset, while the largest experiments used additional images from a Microsoft-hosted archive. Although the validation and test sets remained unchanged, differences between the source datasets may have influenced model performance.

Duplicate-image verification. Exact file duplicates were excluded from the larger training dataset using SHA-256 hashes. However, additional verification for visually identical images with different encodings was not completed. Some undetected overlap between training and evaluation images therefore remains possible.

Training randomness. Each primary configuration was trained once. Model initialization, random augmentation, and the order of training examples may influence results. Multiple training runs would provide a more reliable estimate of average performance and variability.

Model optimization. The comparison used one principal scratch architecture and one pretrained VGG16 architecture. Other network designs, hyperparameters, training durations, or fine-tuning methods could produce different results.

Model selection and evaluation. Validation loss was used to identify the preferred epoch. Test-set performance was also inspected during the sequential experiments, so the reported test results should be interpreted as outcomes of an exploratory comparison rather than an entirely untouched final evaluation.

Computational efficiency. Although training epochs were recorded, total training time, memory consumption, and GPU utilization were not systematically measured. Therefore, conclusions about computational efficiency are based on the training approach and number of epochs, not direct resource measurements.

These limitations do not eliminate the value of the comparison, but they indicate that the exact performance differences should not automatically be expected in other datasets or repeated experiments.

6. Conclusion

The purpose of this assignment was to examine how training sample size and network selection affect the performance of convolutional neural networks.

The results demonstrated that increasing training sample size improved classification accuracy for CNNs trained from scratch. Test accuracy increased from 68.60% with 1,000 training images to 83.00% with 9,996 images.

However, the pretrained VGG16 network achieved substantially better performance across every training sample size. With only 1,000 training images, VGG16 achieved 96.80% test accuracy, outperforming even the largest scratch CNN.

The highest accuracy of the six primary experiments was obtained using pretrained VGG16 with 9,996 training images, which achieved 98.60% test accuracy.

The central conclusion is that **both sample size and network choice influence classification performance, but pretrained networks can substantially reduce the amount of task-specific training data needed to obtain strong predictions.**

For image classification problems with limited labeled data, a pretrained network may therefore be a more effective starting point than training a convolutional network entirely from scratch. When sufficient training data and computational resources are available, training from scratch remains a viable approach, although its relative benefits depend on the application.

Overall, the experiments reinforced the value of transfer learning and demonstrated that increasing data availability and reusing previously learned representations are complementary strategies for improving deep learning performance.

References

Chollet, F. (2021). *Deep learning with Python* (2nd ed.). Manning Publications.

Chollet, F. (n.d.). *Deep learning with Python notebooks: Second edition, Chapter 8, introduction to deep learning for computer vision* [Jupyter notebook]. GitHub. https://github.com/fchollet/deep-learning-with-python-notebooks/blob/master/second_edition/chapter08_intro-to-dl-for-computer-vision.ipynb

Microsoft. (n.d.). *Cats and Dogs image dataset* [Dataset]. Microsoft Download Center. https://download.microsoft.com/download/3/E/1/3E1C3F21-ECDB-4869-8368-6DEBA77B919F/kagglecatsanddogs_5340.zip